Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/28/2005
PI Name:	Landau, David Paul Ph.D.		
Project Title:	Computer Simulations of Confined Quantum Systems in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	30602	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	04/01/2004	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	GSN 4515		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		
Task Description:	The objective is to use computer simulations of appropriate models to improve our understanding of the behavior of quantum systems in confined geometries as the phase transition is approached. Our immediate objective is to use the XY-model of "spins" to describe the critical behavior of superfluid helium. This is accomplished by an appropriate renaming of variables that yields a physically relevant description of the lambda point. We are developing improved methods for carrying out Monte Carlo simulations for systems with two parallel surfaces that play the role of the confining walls. Then, new spin dynamics methods will be used to study the time dependent behavior of the model and thus the thermal conductivity. Data will be compared with those from the BEST and DYNAMX experiments. Path integral Monte Carlo methods are also to be developed to study the behavior of confined Boson systems in different types of fields. Both the techniques developed and the results appropriate to helium may play an important role in the		

	elucidation of behavior of other confined quantum systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Thermal conductivities of pseudo-two dimensional systems of liquid Helium have been determined by computer simulations. The variation was determined as a function of aspect ratio and overall size of the simulation cell. The results show good agreement with experimental data on liquid Helium confined in thin, almost two dimensional chambers. Further development has been made on path integral Quantum Monte Carlo methods and the resultant simulations for weakly interacting hard-core Bosons have been completed for a range of system sizes and scattering lengths and for number densities than are smaller than those used in any previous study. Finite size scaling confirms the consistency of the estimates for the transition temperatures. The resulting values for the shift in the transition temperature are substantially different than those yielded by earlier simulations. They are, however, consistent with theoretical predictions for the leading order shifts but find quite different values for corrections terms. Path integral Monte Carlo simulations have begun for Bose gases in highly elongated traps. Our goal is to explore the change from the mean-field regime to the Tonks-Girardeau gas. To further extend the investigation to realistic gases, dipole interactions have been added to the Hamiltonian, and simulations have begun on the resultant Hamiltonian.
Bibliography Type:	Description: (Last Updated: 02/26/2014)
Articles in Peer-reviewed Journals	Nho K, Landau DP. "Bose-Einstein Condensation Temperature of a Homogenous Weakly Interacting Bose Gas: PIMC Study." Phys Rev A. 2005, in press as of Jan 2005. , Jan-2005